SE 423 Mechatronics

Homework Assignment #Code Commenting

All homework code is expected to be commented and have your initialed on each of the lines.
Points WILL be taken off if this is not done.

Exercise 1: Internal commenting

Below are examples of how we are expecting you to comment your code. **MAKE SURE YOU DO THIS!!!** Also make sure you put in the **top of the file** what the initials to be checked for are!

```
// Andy Jones is commenting this Code. Comments marked by my initials AAJ
//###########################################################################
// FILE: LABstarter_main.c
//
// TITLE: Lab Starter
//###########################################################################
#include <stdio.h>
#include <stdlib.h>
.... Omitted Code
.... You should not omit code in your submission, it has just been removed here for the sake of conciseness
....
uint32_t numTimer0calls = 0;
uint32_t numSWIcalls = 0;
extern uint32_t numRXA;
uint16_t UARTPrint = 0;
uint16_t LEDdisplaynum = 0;
float Kp = 3.4; // AAJ Kp and Ki gains used for the PI controller. Hand tuned for desired response.
float Ki = 29.3; // I did find that a number of gain combinations worked, but settled on these gains.
float Vref = 2.5;
float Vactual = 0;
float Ik = 0;
float Vout = 0;
void main(void)
{
    // PLL, WatchDog, enable Peripheral Clocks
    // This example function is found in the F2837xD_SysCtrl.c file.
    InitSysCtrl();
    .... Omitted Code
    .... You should not omit code in your submission, it has just been removed here for the sake of conciseness
    ....
    // Configure CPU-Timer 0, 1, and 2 to interrupt every second:
    // 200MHz CPU Freq, 1 second Period (in uSeconds)
    ConfigCpuTimer(&CpuTimer0, 200, 5000); //AAJ Set CPU Timer 0 to call its interrupt function every 5ms.
                                           // This is the specified sample rate for the control loop calculations.
    ConfigCpuTimer(&CpuTimer1, 200, 20000);
    ConfigCpuTimer(&CpuTimer2, 200, 40000);
    .... Omitted Code
    .... You should not omit code in your submission, it has just been removed here for the sake of conciseness
    ....
    // Enable global Interrupts and higher priority real-time debug events
    EINT; // Enable Global interrupt INTM
    ERTM; // Enable Global realtime interrupt DBGM

    !!!! There is a second page !!!!!
```

```
    // IDLE loop. Just sit and loop forever
    while(1)
    {
        if (UARTPrint == 1 ) {
            // AAJ Here I am using GPIO67 to see how much time the serial_printf function takes to run.
            // In lab I will connect the Oscilloscope to GPIO67's pin and measure the width of the pulse created by setting
            // high (3.3Volts) before the function is called and then setting GPIO67 low (0Volts) after the function is co
            // GPIO67 is controlled by the GPC registers. GPIO67 is the 4th bit in the GPC registers. Bit 0 is for GPIO64
            // is for GPIO65, Bit 2 is for GPIO66, Bit 3 is for GPIO67, Bit 4 is for GPIO68 and so on. The below bitfield
            // statement hides the bit number from me and allows me to just set the associated bit pertaining to GPIO67. S
            // a bit to 1 in the GPCSET register causes that associated output pin to be high (3.3 Volts). Setting a bit t
            //in the GPCSET register does nothing.
            GpioDataRegs.GPCSET.bit.GPIO67 = 1;
            //AAJ Printing Desired and Actual Speed %.3f tells printf to print 3 decimal places of float variable
            serial_printf(&SerialA,"%.3f Desired Speed, %.3f Actual Speed\r\n",Vref,Vactual);
            UARTPrint = 0;
            // AAJ Set GPIO67 to low (0Volts) so Oscilloscope and see when serial_printf function is complete
            // Setting a bit to 1 in the GPCCLEAR register causes that associated output pin to be low (0 Volts).
            //Setting a bit to 0 in the GPCCLEAR register does nothing.
            GpioDataRegs.GPCCLEAR.bit.GPIO67 = 1
        }
    }
}
// cpu_timer0_isr - CPU Timer0 ISR
__interrupt void cpu_timer0_isr(void)
{
    Vactual = readfeedback(); // AAJ this is a dummy function for this example.
    // AAJ Here I am calculating the PI control law. I am using the "forward rule" to approximate the integral
    // Then I calculate the control effort Vout which is Kp times the error between desired speed and actual speed
    // and Ki times the integral appoximation of the error. I also check for integral windup and then saturate
    // the output between -10 and +10.
    Ik = Ikold + (Vref-Vactual)*0.005;
    Vout = Kp*(Vref-Vactual) + Ki*Ik;
    if (fabs(Vout) > 10) Ik = Ikold;
    if (Vout > 10) Vout = 10;
    if (Vout < -10) Vout = -10;
    outPWM(Vout); //AAJ dummy function to output control
    Ikold = Ik; //AAJ save past state for next 5ms
    CpuTimer0.InterruptCount++;
    numTimer0calls++;
    if ((numTimer0calls%20) == 0) {
        UARTPrint = 1; //AAJ Print to Tera Term every 100ms 5ms * 20 = 100ms
        // Blink LaunchPad Red LED
        GpioDataRegs.GPBTOGGLE.bit.GPIO34 = 1;
    }

    // Acknowledge this interrupt to receive more interrupts from group 1
    PieCtrlRegs.PIEACK.all = PIEACK_GROUP1;
}
.... Omitted Code
.... You should not omit code in your submission, it has just been removed here for the sake of conciseness
....
```

Exercise 2: Function documentation

When you specifically create a function (you do not need to do this for the provided interrupts or the main function, but you are welcome to) you should be using be using Doxygen JAVADOC styling for function documentation (it is what the industry uses so use it!).

Each function should have the following structed documentation:

1. Brief summary (first sentence)
2. A more detailed description.
3. `@param <parameter name> <description of parameter>` for each parameter, ordered by when it appears in the function's parameters
4. `@return <description of what gets returned>`, when applicable (`void` functions would not have this)
5. `@pre <parameter expectations>`, providing expectations of what the input is supposed to satisfy when inputted into it
6. `@post <return expectations>`, providing expectations of how the parameters are modified during the function
7. `@note` or `@warning` when relevant

```c
/**
 * AAJ: A brief history of JavaDoc-style (C-style) comments.
 *
 * This is the typical JavaDoc-style C-style comment. It starts with two
 * asterisks.
 *
 * @param theory Even if there is only one possible unified theory. it is just a
 *               set of rules and equations.
 */
void cstyle( int theory );
```

```c
/**
 * AAJ: Computes the arithmetic mean of an array of doubles.
 *
 * @param data Pointer to the first element of the array.
 * @param length Number of elements in the array (must be > 0).
 * @return The arithmetic mean of the values in the array.
 *
 * @pre data != NULL
 * @pre length > 0
 * @post The input array is not modified.
 */
double mean(const double *data, size_t length);
```

```c
/**
 * AAJ: Opens a file and reads its contents into a buffer.
 *
 * The function allocates memory dynamically for the buffer.
 * The caller is responsible for freeing it with free().
 *
 * @param path Null-terminated string specifying the file path.
 * @param buffer Output pointer that will receive allocated memory.
 * @param size Output parameter storing the number of bytes read.
```

```
 *
 * @return 0 on success, negative error code on failure.
 *
 * @retval 0 Success.
 * @retval -1 File could not be opened.
 * @retval -2 Memory allocation failed.
 *
 * @pre path != NULL
 * @pre buffer != NULL
 * @pre size != NULL
 * @post On success, *buffer points to valid allocated memory.
 *
 * @warning The function does not append a null terminator.
 */
int read_file(const char *path, unsigned char **buffer, size_t *size);
```

```
/**
 * AAJ: Initializes the ADC peripheral.
 *
 * Configures hardware registers and enables interrupts required
 * for analog-to-digital conversion.
 *
 * @param config Pointer to ADC configuration structure.
 *
 * @return true if initialization succeeds, false otherwise.
 *
 * @pre config != NULL
 * @post ADC hardware is ready for sampling.
 *
 * @note This function must be called before adc_read().
 */
bool adc_init(const adc_config_t *config);
```

Exercise 3: Function template

Below is a template that you can use,

```
/**
 * AAJ: Brief one-line summary.
 *
 * Detailed description if needed.
 *
 * @param name Description.
 * @return Description.
 *
 * @pre Conditions that must hold.
 * @post Guarantees after execution.
 */
```